

SAFETY DATA SHEET

Section 1: PRODUCT AND COMPANY IDENTIFICATION

1.1. Product identifier

Product name BRITE-NICKEL BNI-600

1.2. Other means of identification

None

1.3. Recommended use of the chemical and restrictions on use:

Recommended Use Bright nickel plating additive

1.4. Details of the supplier of the safety data sheet

Address A Brite Company
3217 Wood Drive
Garland, TX 75041

Phone number 1-888-8ABRITE

Website www.abrite.com

1.5. Emergency phone number

Emergency telephone 1-800-424-9300 (CHEMTREC)

Section 2: HAZARDS IDENTIFICATION

2.1. Classification of the chemical in accordance with 29 CFR 1910 (OSHA HCS)

ACUTE TOXICITY (oral)	Category 4
ACUTE TOXICITY (dermal)	Category 4
SKIN CORROSION/IRRITATION	Category 2
SERIOUS EYE DAMAGE/ EYE IRRITATION	Category 2A
SKIN SENSITIZATION	Category 1

2.2. Label Elements



Pictogram

Signal Word Warning

Hazard Statements

Harmful if swallowed or in contact with skin.
Causes serious eye irritation.
Causes skin irritation.
May cause an allergic skin reaction.

Precautionary statements

Prevention

Wear protective gloves. Wear eye or face protection. Wear protective clothing. Avoid breathing vapor. Do not eat, drink or smoke when using this product. Wash hands thoroughly after handling. Contaminated work clothing should not be allowed out of the workplace.

<u>Response</u>	IF SWALLOWED: Call a POISON CENTER or physician if you feel unwell. Rinse mouth. IF ON SKIN: Wash with plenty of soap and water. Call a POISON CENTER or physician if you feel unwell. Take off contaminated clothing. Wash contaminated clothing before reuse. If skin irritation or rash occurs: Get medical attention. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical attention.
<u>Storage</u>	Store in cool/well-ventilated place. Keep container tightly closed
<u>Disposal</u>	Dispose of contents and container in accordance with local, regional, national, and international regulations.

2.3. Hazards not otherwise classified

None known

2.4. Ingredient(s) with unknown acute toxicity

None known

Section 3: COMPOSITION/INFORMATION ON INGREDIENTS**3.1. Chemical name and concentration/concentration ranges**

Ingredient name	%	CAS Number
Proprietary Additive(s)	1-10	-
Proprietary Alcohol	1-10	-
prop-2-yn-1-ol	1-10	107-19-7
Glycol	0.1-1.0	-
Inorganic acid	0.1-1.0	-

3.2. Common name and synonyms

See 3.1

3.3. CAS number and other unique identifiers

See 3.1

3.4. Impurities/stabilizing additives

See 3.1

Section 4: FIRST AID MEASURES**4.1. Description of first aid measures**

Skin contact	Wash with plenty of soap and water. Remove contaminated clothing and shoes. Wash contaminated clothing thoroughly with water before removing it, or wear gloves. Continue to rinse for at least 15 minutes. Get medical attention. If necessary, call a poison center or physician. In the event of any complaints or symptoms, avoid further exposure. Wash clothing before reuse. Clean shoes thoroughly before reuse.
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Ingestion

Wash out mouth with water. Remove dentures if any. Remove victim to fresh air and keep at rest in a position comfortable for breathing. If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Stop if the exposed person feels sick as vomiting may be dangerous. Do not induce vomiting unless directed to do so by medical personnel. If vomiting occurs, the head should be kept low so that vomit does not enter the lungs. Get medical attention. If necessary, call a poison center or physician. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband.

Inhalation

Remove victim to fresh air and keep at rest in a position comfortable for breathing. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Get medical attention if adverse health effects persist or are severe. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband. In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours.

4.2. Most important symptoms/effects, acute and delayed

Causes serious eye irritation. No known significant effects or critical hazards. Harmful in contact with skin. Causes skin irritation. May cause an allergic skin reaction. Harmful if swallowed.

4.3. Indication of immediate medical attention and special treatment, if necessary

In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours. No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

Section 5: FIRE FIGHTING MEASURES**5.1. Suitable/ unsuitable extinguishing media**

Use extinguishing agent suitable to surrounding fire.

5.2. Specific hazards arising from chemical

In a fire or if heated, pressure will occur and the container may burst. Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
nitrogen oxides
sulfur oxides

metal oxide/oxides

5.3. Special protective equipment and precautions for fire-fighters

Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

Section 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment, and emergency procedures

No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilled material. Avoid breathing vapor or mist. Provide adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Put on appropriate personal protective equipment. If specialised clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials. See also the information in "For nonemergency personnel". Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

6.2. Methods and materials for containment and cleaning up

Small spill: Stop leak if without risk. Move containers from spill area. Dilute with water and mop up if water-soluble. Alternatively, or if water-insoluble, absorb with an inert dry material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor. Large spill: Stop leak if without risk. Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations (see Section 13). Dispose of via a licensed waste disposal contractor. Contaminated absorbent material may pose the same hazard as the spilled product. Note: see Section 1 for emergency contact information and Section 13 for waste disposal.

Section 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Put on appropriate personal protective equipment (see Section 8). Persons with a history of skin sensitization problems should not be employed in any process in which this product is used. Do not get in eyes or on skin or clothing. Do not ingest. Avoid breathing vapor or mist. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Empty containers retain product residue and can be hazardous. Do not reuse container.

Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.

7.2. Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10)

and food and drink. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabeled containers. Use appropriate containment to avoid environmental contamination.

Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Ingredient Name	Exposure limits
prop-2-yn-1-ol	ACGIH TLV (United States, 4/2014). Absorbed through skin. TWA: 2.3 mg/m ³ 8 hours. TWA: 1 ppm 8 hours.
Inorganic acid	NIOSH REL (United States, 10/2013). Absorbed through skin. TWA: 2 mg/m ³ 10 hours. TWA: 1 ppm 10 hours. ACGIH TLV (United States, 4/2014). Notes: Refers to Appendix A -- Carcinogens. Inhalable fraction. See Appendix C, paragraph A. Inhalable Particulate Mass TLVs (IPM-TLVs) for those materials that are hazardous when deposited anywhere in the respiratory tract. ACGIH 2005 Adoption STEL: 6 mg/m ³ 15 minutes. Form: Inhalable fraction TWA: 2 mg/m ³ 8 hours. Form: Inhalable fraction

8.2. Appropriate engineering controls.

Good general ventilation should be sufficient to control worker exposure to airborne contaminants. Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

8.3. Individual protection measures

Eye/face protection

Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Contaminated work clothing should not be allowed out of the workplace. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Skin/body protection

Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary. Considering the parameters specified by the glove manufacturer, check during use that the gloves are still retaining their protective properties. It should be noted that the time to breakthrough for any glove material may be different for different glove manufacturers. In the case of mixtures, consisting of several substances, the protection time of the gloves cannot be accurately estimated. Personal protective equipment for the body

should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product. Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.

Respiratory protection

Use a properly fitted, air-purifying or air-fed respirator complying with an approved standard if a risk assessment indicates this is necessary. Respirator selection must be based on known or anticipated exposure levels, the hazards of the product and the safe working limits of the selected respirator.

General considerations

Always wash hands before smoking/eating/drinking or using the toilet. Wash contaminated clothing and other PPE before storage or reuse. Eyewash fountains and safety showers must be easily accessible and close to the workstation location.

Section 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Appearance	Clear amber liquid
Odor	Slight
Odor threshold	No applicable information is available
pH	No applicable information is available
Melting/freezing point	No applicable information is available
Boiling point/boiling range	100°C (212°F)
Flash point	No applicable information is available
Evaporation rate	No applicable information is available
Flammability (solid/gas)	No applicable information is available
Flammability limits	No applicable information is available
Vapor pressure	No applicable information is available
Vapor density	No applicable information is available
Relative density	1.04
Solubility(ies)	Easily soluble in cold or hot water
Partition coefficient (n-octanol/water)	No applicable information is available
Auto-ignition temperature	No applicable information is available
Decomposition temp	No applicable information is available
Viscosity	No applicable information is available

Section 10: STABILITY AND REACTIVITY

10.1. Reactivity

No specific test data related to reactivity available for this product or its ingredients.

10.2. Chemical stability

The product is stable.

10.3. Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.

10.4. Conditions to avoid

Excessive heat

10.5. Incompatible materials

Reactive or incompatible with the following materials: oxidizing materials, metals, acids and alkalis.

10.6. Hazardous decomposition products

Under normal conditions of storage and use, hazardous decomposition products should not be produced. Under normal conditions of storage and use, hazardous polymerization will not occur.

Section 11: TOXICOLOGICAL INFORMATION

11.1. Information on the likely routes of exposure and symptoms

Product/ingredient name	Result	Species	Dose	Exposure
Proprietary Additive(s)	LD50 Oral	Rat	270 mg/kg	-
Proprietary Alcohol	LD50 Oral	Rat	>500 mg/kg	-
prop-2-yn-1-ol	LC50 Inhalation Gas.	Rat	1040 ppm	1 hours
	LC50 Inhalation Vapor	Rat	0.75 mg/l	4 hours
	LD50 Dermal	Rabbit	16 mg/kg	-
	LD50 Dermal	Rabbit	88 mg/kg	-
	LD50 Oral	Rat	20 mg/kg	-
Glycol	LD50 Oral	Rat	100 mg/kg	-
Inorganic acid	LD50 Oral	Mouse	3450 mg/kg	-
	LD50 Oral	Rat	2500 mg/kg	-
	LD50 Oral	Rat	2660 mg/kg	-
	LDLo Dermal	Child	1500 mg/kg	-
	LDLo Dermal	Infant	1200 mg/kg	-
	LDLo Dermal	Man	2430 mg/kg	-
	LDLo Oral	Human	214.28 mg/kg	-
	LDLo Oral	Rat	3000 mg/kg	-
	LDLo Oral	Woman	200 mg/kg	-

Irritation/Corrosion

Product/ingredient name	Result	Species	Exposure
Inorganic acid	Skin – Mild irritant	Human	72 hours 15 milligrams intermittent

11.2. Numerical measures of toxicity

Route	ATE
Oral	1387.6 mg/kg
Dermal	1478.2 mg/kg
Inhalations (gasses)	48042.7 ppm
Inhalation (vapors)	69.29 mg/l

11.3. Carcinogenicity

No applicable toxicity data

Section 12: ECOLOGICAL INFORMATION

12.1. Ecotoxicity

Product/ingredient name	Result	Species	Exposure
Proprietary Alcohol	Acute LC50 >100 mg/l Fresh water	Fish	96 hours
	Acute LC50 >1000 mg/l	Micro-organism	17 hours
prop-2-yn-1-ol	Acute LC50 1.44 mg/l Fresh water	Fish – Pimephales promelas	96 hours
Glycol	Acute LC50 53600 to 58300 µg/l Fresh water	Fish - Pimephales promelas - Juvenile (Fledgling, Hatchling, Weanling)	96 hours
Inorganic Acid	Acute LC50 45.5 mg/l Fresh water	Crustaceans – Ceriodaphnia dubia	48 hours
	Acute LC50 133000 µg/l Fresh water	Daphnia - Daphnia magna – Neonate	48 hours
	Acute LC50 75 mg/l Marine water	Fish - Pagrus major	96 hours
	Chronic NOEC 6000 µg/l Fresh water	Daphnia - Daphnia magna	21 days
	Chronic NOEC 2100 µg/l Fresh water	Fish - Oncorhynchus mykiss	87 days

12.2. Persistence/degradability 90 to 100% - readily – 21 days

12.2. Bioaccumulative potential Low

12.3. Mobility in soil No applicable information is available

12.4. Other adverse effects No known significant effects or critical hazards

Section 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers

Section 14: TRANSPORT INFORMATION

UN number	Not regulated under 49 CFR 172.101.
UN proper shipping name	No applicable information is available
Transport hazard class(es)	No applicable information is available
Packing group, if applicable	No applicable information is available

Environmental hazards	No applicable information is available
Transport in bulk	No applicable information is available
Special precautions	No applicable information is available

Section 15: REGULATORY INFORMATION

15.1. Regulatory information

Product classified according to OSHA CFR 29 1910.1200. See section 2.

Toxic Substances Control Act (TSCA)

Chemical ingredients are on the TSCA inventory.

SARA Title III Section 302

No ingredients subject to reporting requirements

SARA Title III Section 313

prop-2-yn-1-ol, 107-19-7, 1-10%

SARA Title III Section 311/312

Immediate (acute) health hazard

Massachusetts/Pennsylvania/New Jersey Right to Know Components

No products were found.

California Prop. 65 Components

WARNING: This product contains less than 0.1% of a chemical known to the State of California to cause cancer.

RQ

No products were found.

Section 16: OTHER INFORMATION

HMIS Hazard Codes (minimal = 0, slight = 1, moderate = 2, serious = 3, severe = 4)

Health: 1

Fire: 0

Reactivity: 0

Special: NA

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Disclaimer

To the best of our knowledge, the information contained herein is accurate. However, neither A Brite Company nor any of its employees or subsidiaries assume any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

End of Safety Data Sheet